BHARAT MISHRA

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EDUCATION

Bachelor of Technology, VIT Bhopal University, Computer Science and Engineering **CGPA: 9.11** 2027
Grade XII, Vidyatree Modern World School **Percentage: 95.25** 2023

TECHNICAL SKILLS

- **Programming Languages:** C/C++, Python, SQL
- **ML/AI:** TensorFlow, Scikit-learn, Pandas, FastAPI, NumPy, OpenAI Gym, OpenCV, MediaPipe, Pydantic
- **Developer tools:** GitHub, Git, Docker, MySQL, PostgreSQL
- **Core concepts:** Deep learning, Embeddings, Reinforcement Learning, LLMs, Agentic AI, Computer Vision

PROJECTS

DepoIndex: AI Legal Document Indexing *LLM APIs, Prompt Engineering, PDF Processing*
github.com/Bharat269/DepoIndex

- Engineered an AI-powered tool to automate the generation of a table of contents from legal deposition PDFs using an efficient preprocessing pipeline that annotates text with precise page and line numbers, **minimizing prompt token** count while enabling the LLM to extract topics with accurate source references.
- Achieved **95% page-level and 85% line-leve**l accuracy in topic referencing, delivering a highly reliable tool for rapid automated document indexing.

Attentia: Adaptive Learning for Neurodivergent Children *FastAPI, OpenCV, MediaPipe, Q-learning*
github.com/Bharat269/attentia.ai

- Designed an adaptive learning system that detects attention/emotional state and performs tailored interventions (e.g., adjusting course difficulty, soothing with music, reducing distraction via animations). Built the training environment for the Q-learning engine using a mix of deterministic and stochastic rules to simulate user behaviour.
- Engineered **multi-modal attention detection pipelines** (facial landmarks, head pose, audio signals) with personalized baseline calibration, using a persistent singleton architecture for efficient and accurate real-time engagement tracking and low-latency FastAPI integration.

MovieLens: Semantic Movie Search *FastAPI, Docker, TensorFlow, Embeddings, Recommendation Systems*
github.com/Bharat269/MovieMix

- Deployed a production-grade semantic search system with a **under 50ms** (improved from 2 seconds) FastAPI+Docker inference pipeline to convert raw user input tags into 100-D embeddings, and retrieve movies via vectorized cosine similarity against movie embeddings derived from **matrix factorization**.
- Engineered a 500-token tag vocabulary and a hypersphere-normalized text-to-vector neural network for **solving the hubness problem** and also solving the cold-start problem by generating embeddings from tags alone achieving **0.52 cosine similarity**.

Rescue: Environment Navigation *Multi-Agent Reinforcement Learning, Custom Environment*
github.com/Bharat269/ProjectExhibition-2-Search-Rescue

- Developed autonomous agents using epsilon-greedy Q-learning within a custom-built multi-agent physics environment, optimizing pathfinding in dynamic, natural disaster simulations and **reducing search time by 90%** compared to a random-walk baseline. Simulated human behaviour with Social Forces Movement theory.

ACHIEVEMENTS AND CERTIFICATIONS

- **Generative AI training by Kaggle and Google (2024)**: Transformers, LLMs, and prompt engineering.
- **Global Rank 8261, TCS CodeVita (2024):** Solved advanced algorithmic problems.